



R&D collaboration, social coordination, and standardization: evidence from the Chinese automotive industry

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Abstract

This study examines the role of research and development (R&D) collaboration in the race to set standards in the Chinese automotive industry. We unravel how different properties of R&D collaboration provide firms with social capital that contributes to their involvement in formal standard setting. To test our hypotheses, we use a sample of 508 firms involved in collaborative R&D and standardization from 1987 to 2010. Our results show that firms with stronger and more efficient connections to R&D partners are more likely to set (domestic) standards. Furthermore, firms are more likely to set collaborative standards as a principal editor or non-collaborative standards when they possess stronger connections to their R&D partners. This relationship becomes more pronounced when the proportion of Chinese state-owned enterprises in the firm's R&D network increases. These findings provide valuable insights for both managers and policymakers, as collaborative partnerships are primary strategic assets that can be leveraged for lobbying and hence, drive competitive advantage in standard competition.

Keywords R&D collaboration · Standardization · Social coordination · China · Automotive industry

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1 Introduction

The race to define an industry dominant design has long been deemed to be ultimately decided by customer response. A customer installed base that is reinforced by direct and indirect network externalities is a key source of power in industrial standardization (Farrell & Saloner, 1985; Katz & Shapiro, 1985). This demand-based approach emphasizes the effect of spontaneous coordination and the importance of market-based mechanisms in the standard setting process (Dew & Read, 2007; Leiponen, 2008). However, as Dosi (1982) notes, industrial technological change is driven by a mixture of demand-pull and technology-push. Research therefore increasingly focuses on standards that emerge from supportive firms involved in the co-creation of new technologies (Gallagher, 2012; Gao, 2015; Johansson et al., 2019; Wen et al., 2020) and, consequently, on standard competition among coalitions with diverse technological orientations (Bar & Leiponen, 2014; Blind & Mangelsdorf, 2016; Lichtenthaler, 2012; Soh, 2010; Wang & Xie, 2011).

Although these latest advances indicate that the competition for industry standards involves forming technology-oriented coalitions among firms and other industry actors, they generally tend to consider coalitions as exogenous or do not tie the properties of collaborative activities to the actual setting of standards (Wen et al., 2020). Notable exceptions are studies by Bar and Leiponen (2014), Dai et al. (2018) and Wen et al. (2020), who show that firms in the telecom and automotive industry respectively leverage their social network of former standard alliance partners to support new technological specifications within a standard setting organization (SSO). While these microlevel studies address the roles of social capital and network connectivity in standard-setting committees, they do not unravel whether the technology-push from interorganizational connections outside the SSO alliance network also facilitates standardization in a competitive environment. To improve our understanding of this still unexplored mechanism, this study examines the relationship between interorganizational cooperation in research and development (R&D) and formal standardization.

Leveraging insights from social capital and transaction costs theory, we argue that social coordination of R&D partners is a predominant source of competitive advantage in standard competition. Interorganizational network connections provide firms with social capital in the forms of knowledge or control over external resources (Owen-Smith & Powell, 2004; Zaheer & Bell, 2005). In particular, social capital also arises from relational and structural embeddedness in R&D networks (Ardila et al., 2020; Schilling & Phelps, 2007; Xu & Zeng, 2021; Xu et al., 2017). However, this strand of literature has not attempted to explain how such embeddedness in R&D networks plays a role in standardization. Hence, we examine how different properties of interorganizational connections, measured using granted co-patents resulting from R&D collaboration, contribute to firms' involvement in formal or de jure standard setting. In addition, extant research has generally overlooked the heterogeneity of standards regarding their technological intensity, market value, and the varying role of different stakeholders' contributions to the drafting process (Blind & Gauch, 2009; Dai et al., 2018; Kwak et al., 2011). Therefore, we also assess whether the impact of interorganizational network connections on standardization varies depending on the type of standard and whether (connections with) potentially dominant firms within the network increase the propensity to set standards.

A particularly suitable context for examining our research questions is the Chinese automotive industry. This industry has for long been considered by the Chinese state authorities as essential for technological development and industrial prominence. The sheer size of the

Chinese automotive market and the intense competition makes that firms rely on intensive cooperation and coordination to constantly adapt existing product designs (Zhao & Anand, 2009). Furthermore, the complexity of the production and supplier networks and the role of Chinese state-owned enterprises further emphasize the importance of collaboration, social coordination, and standardization among all stakeholders involved in the production process (Dai et al., 2018; Wen et al., 2020). Consequently, the industry was among the first in China to launch systematic standardization through formal coordination procedures.

To address our research question, we analyze an unbalanced panel dataset of 508 firms involved in collaborative R&D networks and standardization in the Chinese automotive industry from 1987 to 2010. The dataset includes a diverse group of firms, ranging from relatively small and medium-sized car part manufacturers to internationally well-known multinationals like Dongfeng motor corporation and First Automobile Works Group. Using panel Poisson fixed effects models, our estimations show that strong relationships among R&D network partners are a source of competitive advantage in standard setting. Firms acting as the (principal) editor of a standard, particularly benefit from network connections when the proportion of Chinese state-owned enterprises in their R&D network increases. While our main results provide only partial support for our hypotheses, we obtain additional support in further analyses examining subsets of the data.

We contribute to the literature by providing an understanding of the still unexplored link between the intensity and structure of R&D collaborations for setting standards. As an additional contribution, we offer new insights into the varying relevance of R&D collaborations for setting different types of standards. Potential endogeneity concerns over the direction of the estimated relationships are ruled out based on the outcomes of our Granger causality estimations.

2 Background and related literature

2.1 Formal standard setting through standard setting organizations

Technology standards are defined as specifications or protocols that determine the compatibility of products in a network of users (Gallagher, 2012; Murphree & Breznitz, 2018; Shapiro & Varian, 1999). Under conditions of technological progress, standards are essential to ensure focus and conformity to create new markets with larger economies of scale and increase the ability to develop and sell complementary products and processes (Chiao et al., 2007). Notwithstanding their usefulness, standards also lock in markets to specific technologies (Leiponen, 2008). Firms that are able to push their proprietary technology as the industry standard can eliminate competition, reduce the incentives to innovate, and often hamper market evolution for extended periods of time (Bekkers et al., 2002). The promotion of specific proprietary technologies as an industry standard may therefore be particularly lucrative and, consequently, provoking firms' strategic behavior at early stages of the standardization process.

Standards can emerge in different ways. De facto standards are dominant designs that arise naturally from widespread public adoption or market-based mechanisms without any formal approval (Chiao et al., 2007). Typically used examples are the customer gravitation towards the use of the QWERTY keyboard or the Blu-ray digital versatile disc (DVD) respectively (Gallagher, 2012). For the automotive industry, well-known examples are Ford's model T as the dominant car design in the early twentieth century or the long

(but now ending) use of the internal combustion engine as sole power source (Fujimoto, 2014). In contrast, *de jure* standards generally follow a consensus-based process of formal approval and administration by an authoritative (inter)national organization (Backhouse et al., 2006; Blind & Mangelsdorf, 2016). Formal standards thus have greater legitimacy and can be used as legal requirements in technical specifications (Leiponen, 2008). Computer hardware standards such as HDMI or USB, or the standards of the Internet Engineering Task Force that govern internet infrastructure (Simcoe, 2014) are examples of formal standards that people use on a daily basis.¹ Many car safety requirements and devices also include formal standards such as exhaust emissions, airbags, and (anti-lock) braking systems.

Apart from *de facto* standards (Lichtenthaler, 2012), formal standardization is an important mechanism for defining technology adoption (Chiao et al., 2007; Farrell & Saloner, 1985). Establishment of formal standards relies on social coordination mechanisms that usually involve the operation of SSOs. SSOs provide fora where stakeholders discuss and coordinate ideas to achieve consensus in technology choices that are collectively promoted as industrial standards (Simcoe, 2014). Normally, a committee to oversee the process of drafting standards is formed of members that represent different stakeholders.² Furthermore, SSO members can propose standards or ask for amendments to existing ones. The content of these proposals is subject to review by other stakeholders and committee members, who may critique them and require revision. Consequently, SSOs entail high levels of negotiation, coordination, and compromise along with competition among diverse stakeholders, to achieve standards with acceptable levels of compatibility among complementary products and services (Farrell & Saloner, 1985). As noted by Shapiro and Varian (1999: 240) “the [formal] standard setting process is a wild mix of politics and economics [...]”. Moreover, Backhouse et al., (2006: 414) assert that politics and power “operate silently but relentlessly in the generation and institutionalization of a standard.”

The processes and rules SSOs use to draft and set formal standards drive firms to constantly seek alignment among stakeholders (Simcoe, 2014). For example, in Backhouse et al.’s (2006) study on the international information security standard ISO/IEC 17,799, they report that established relationships and memberships in both social and institutional groupings reflect sources of power that influence the adoption of formal standards. For the global system of mobile telecommunications (GSM) standard case, Bekkers et al. (2002) are more specific and reveal how firms owning essential technologies occupy more central positions in firm alliance networks to define standards. In addition, Bar and Leiponen (2014) show that firms in the telecommunications industry use alliances or actively seek connections to standard committees to lobby and influence the outcomes of standard setting within the Third-Generation Partnership Project SSO. While the literature directs most attention to the role of firms as stakeholders in SSOs, governments may also support or act as a (central) stakeholder in standard setting (Mattli & Büthe, 2003). Government involvement is particularly apparent in emerging markets (Murphree & Breznitz, 2018), where

¹ HDMI (High-definition multimedia interface) is a connector used in systems for high-definition media and USB (Universal serial bus) is a standardized cable and connector protocol for computers and peripheral devices.

² In practice, there is considerable variety in SSO characteristics. For instance, there are SSOs at different geographical levels, including the international, regional, and national level. Furthermore, the nature of SSO membership is often differentiated; some SSOs consist of corporate members only, while others involve individual members or additional interest groups, including research institutes, universities, and government agencies.

setting standards is perceived a powerful strategy for technology catch-up (e.g., Gao, 2015; Kwak et al., 2011).

These examples highlight that formal standardization activities are characterized by social coordination and collective bargaining within SSOs. Hence, the extent to which firms are directly embedded in networks of relationships among industrial stakeholders and committee members is likely to affect the outcomes of formal standard setting procedures. The question remains however whether collaborative activities and social coordination with R&D partners outside these coalitions are also important in standardization.

2.2 Social coordination

R&D networks reveal how firms and other industry actors are technologically embedded in an industry and primarily involve a variety of organizational arrangements among diverse stakeholders. These relationships are established to exchange information and technological knowledge, often across geographic, business, and institutional boundaries, from core actors to peripheral ones and vice versa (Owen-Smith & Powell, 2004; Schilling & Phelps, 2007; Zaheer & Bell, 2005). However, it is not possible to benefit from the advantages of collaboration without social capital. Following Coleman's (1988) theoretical conceptualization, social capital refers to the benefits resulting from a firm's relationship with other actors. The more social capital a firm can leverage, the less transaction costs are involved to manage these relationships (Walker et al., 1997). In a network context, relationships provide a valuable mechanism for building trust, sharing behavioral norms, and establishing mutual expectations that contribute to building social capital among peers (Granovetter, 1985). By perceiving relationships as a social structure, existing social connections can be appropriated for different purposes (Coleman, 1988). For firms involved in formal standardization activities, R&D network connections provide opportunities to accumulate social capital that can be leveraged to promote innovative technologies and align industrial peers in the process of defining a dominant design. Among these opportunities for social coordination, building social capital through common interests, trust, and aligned actions are the most prominent for setting standards.

R&D collaboration requires a degree of interdependence among partners (Gulati & Singh, 1998). Because their resources originate from unique endowments and capabilities with varied historical paths, collaborative partners must manage organizational difficulties, including the "not invented here" syndrome (Hussinger & Wastyn, 2016). To create competitive advantage from cooperation, investments in organizational mechanisms and procedures that demand extensive social coordination efforts are essential (Krishnan et al., 2006). Over time, implementing unified procedures instigates technological and organizational routines, increases interdependency, and tends to confine collaborative partners to a specific technological trajectory (Rycroft & Kash, 2002). Hence, collaborating firms and other industry actors are driven by a common interest and vision to accommodate their shared technological patterns. Establishing common interests and vision thus propels R&D partners to leverage their social capital and support each other in standardization activities.

Trust among partners in R&D collaboration also enhances social capital. As a complex social heuristic, trust involves relationships in which partners do not act at the expense of others for their own benefit (Delhey & Newton, 2005). Trust can be seen as an implicit contract replacing a formal governance structure to attenuate opportunism and, consequently, reduce the transaction costs related to interaction (Hill, 1990; Cummings and Bromily 1996). Forming trust takes time, and frequent interactions that deepen mutual

understanding of each other's knowledge base, technological preferences, and values are essential to improve trust (Uzzi, 1997). In R&D collaboration where knowledge appropriation concerns are preeminent, trust is essential for partners to commit resources (Gulati & Singh, 1998), exchange knowledge, and share responsibilities (Bouty, 2000). Trust is a particularly useful social coordination mechanism for standard setting as it facilitates constructive management of diverging views (Zaheer et al., 1998). Because trust builds on reciprocity between partners, it also reduces uncertainty and encourages mutual support when it comes to achieving consensus in adopting a new technology as an industry standard (Krishnan et al., 2006).

In addition to trust, the success of R&D collaboration is increasingly based on strategically aligned objectives (Duysters & De Man, 2003). Partners involved in an R&D project need to develop a mutual understanding of their technology's market implications (Emden et al., 2006). To increase the probability of success and avoid market fragmentation, collaborative partners must synchronize their goals (Leiponen, 2008) and promote the applicability and usefulness of their new technology (e.g., Bekkers et al., 2002; Soh, 2010). Standardization is facilitated when partners who are determined to set their technology as the industry standard have aligned objectives. Since most SSOs use the consensus rule to set the eventual standard, leveraging social capital in existing connections and lobbying among other industry actors functions as a pre-market standardization mechanism. Partners that strategically align their objectives will find it easier to socially coordinate their actions and justify the supporting-firm base of their shared technology (Backhouse et al., 2006). As a result, they are more likely to set formal standards.

2.3 R&D collaboration and standardization

Since R&D collaboration networks consist of techno-social relationships among industry actors to co-create new technologies, it can be argued that the different properties of these relationships provide various types of social capital (Schilling & Phelps, 2007). Given the importance of social coordination in standardization, these relational characteristics are likely to affect the propensity to set standards. To unravel the impact of a firm's relationships with its R&D collaborative partners on standard setting, we explore three constructs of connectivity that reflect different relational properties: The number of direct ties, tie strength, and the network efficiency of the ties.

2.3.1 Direct connections and the cost of cooperation

Within R&D networks, direct connections between collaborative partners facilitate technology diffusion and establishment of common interests. For instance, Soh (2010) argues that firms that tap into larger groups of industry actors and distribute information about the preferred standard to a wider group of recipients increase awareness among more producers. Furthermore, Dai et al. (2018) demonstrate that firms with expanded business linkages enhance their influence in standard competition. However, to make R&D collaboration a success, firms need to build social capital and learn how to cooperate effectively. Given that building social capital takes time and depends on frequent interactions, commitment, and reciprocity (Coleman, 1988), the transaction costs of R&D cooperation increase when firms start working with new partners (Ring & Van de Ven, 1992; Gulati & Singh, 1998).

Because SSOs use committees with representatives from supporting and stakeholder organizations to reach consensus and set technology standards (Chiao et al., 2007), the

more social capital a firm can leverage in its R&D network, the stronger the support in the standard setting process. Since increasing numbers of direct connections bring additional transaction costs with limited social capital, having more direct connections does not necessarily lead to a larger supporting-firm base. In fact, the impact of having many direct connections is likely to be negative, as establishing new connections can be seen as opportunity cost at the expense of building trust and investing in aligned objectives to make existing partnerships more effective (Bar & Leiponen, 2014). With an increasing number of direct connections, the firm's relationships become more "atomized" and thus, inherently irrelevant (Ring & Van de Ven, 1992). Besides, firms with many connections may also be perceived with caution by SSOs due to their strong bargaining power and the corresponding risk of appropriating the standard setting process (Llanes, 2019; Sidak, 2009). As a result, many SSOs have adopted governance measures to mitigate such risks, including ex-ante disclosure of standard-essential patents, royalty rates, and the implementation of a fair, reasonable, and non-discriminatory commitment policy.

In contrast to enlarging the group of supporting industry actors in de facto standardization, an increasing number of direct connections does not enhance firm competitive positions in formal standardization. Consequently, a focal firm with a relatively large number of direct connections to other inventors is less well positioned to have its voice heard in standard setting. Therefore, we hypothesize:

Hypothesis 1 Firms with more direct connections in their R&D network are less likely to set formal standards.

2.3.2 Strong direct connections and coordination

The nature and strength of R&D collaborative efforts vary by a relationship's duration, interaction frequency, commitment level, and reciprocity. Strong connections facilitate the growth of social capital by building trust and mutual support, which increase the willingness to share detailed proprietary and tacit knowledge (McFadyen et al., 2009; Reagans & McEvily, 2003). This dependency can, for instance, be translated into a firm adopting a shared technology and co-operating to release complementary products. When combined, these socially coordinated actions may contribute to attracting more adopters or an increasing market share of a new technology. Similarly, strong connections in R&D networks support interactions between partners to develop technologies that serve their common interest in standard competition (Bar & Leiponen, 2014). When the connections become stronger, collaborators can be expected to be more committed and mutually interdependent and therefore more likely to support the same dominant design. Consequently, in formal standardization scenarios, strongly connected firms are more likely to ally in standard proposals, coordinate conflicts and constructive solutions, and together deal with criticism raised during the SSO review process. Thus:

Hypothesis 2 Firms with stronger direct connections in their R&D network are more likely to set formal standards.

2.3.3 Network openness and social support

While firms are expected to be reluctant to establish new R&D partnerships due to the cost considerations of building social capital, this does not imply that network expansion cannot

be beneficial (Rost, 2011). A potential downside of continuous investment in stronger relationships is over-embeddedness (McEvily & Marcus, 2005; Uzzi, 1997). The costs of building social capital and involvement of tacit and proprietary knowledge may make the network of R&D collaborative partners cohesive, indicating a closed network, where new knowledge is prevented from reaching the networked firms through outside connections (Soda, 2011). Within R&D collaborative networks, however, connections that offer access to new and nonredundant knowledge are generally good sources for new innovations and business opportunities (Ardila et al., 2020; Fleming et al., 2007; Schilling & Phelps, 2007). Furthermore, as emphasized by Inkpen and Beamish (1997), if a firm bridges key knowledge resources in collaborative activities, it increases its power over other partners. Overall, maintaining exclusive and nonredundant connections to R&D partners provides unique opportunities to accumulate social support for promoting new technologies and coordinating the exchange of information and knowledge.

Network openness and building social support through nonredundant connections are highly relevant for standard setting. Since standards are developed to achieve compatibility among stakeholders, having access to outside and non-redundant network connections offers a unique opportunity to balance technological interests and, simultaneously, align varied objectives among members. In addition, firms with bridging positions in the network also increase their bargaining power because other stakeholders in the network depend on them (Walker et al., 1997; Zaheer & Bell, 2005). Although SSO committees account for the potential detrimental effects of the bargaining power of highly connected firms, they are also aware that a non-redundant network of a well-connected firm encourages inclusive consideration of diverse interests and contributes to an efficient knowledge exchange that enables compromises, resolves conflicts, and steers consensus (Ranganathan et al., 2018). Consequently, to align industry actors, firms with unique outside connections can leverage this dependency and effectively negotiate and bargain with stakeholders to reach collective support for standardizing a new technology. Firms with nonredundant bridging positions are thus likely to be more powerful in standard competition. Hence:

Hypothesis 3 Firms with a higher proportion of nonredundant connections in their R&D network are more likely to set formal standards.

2.4 Chinese state-owned enterprises as network actors

The power of leveraging social capital to influence standard setting outcomes also varies with the characteristics of the stakeholders involved. Alongside firms, the government can act as a predominant stakeholder in standard setting (Mattli & Büthe, 2003). Not only as the authoritative power establishing rules and regulations, but also as the principal of Chinese state-owned enterprises that act as important stakeholders with access to government information, support, and scarce resources (Zhou et al., 2017). Because the quality of public governance differs across countries, also the role of Chinese state-owned enterprises as state agents in standardization differs. In this context China represents a distinct case, as the country develops a standards policy that favors domestic standards, often involving Chinese state-owned enterprises, to promote indigenous innovation and technological independence (Liu & Li, 2016).

In the Chinese standardization system both government control and social coordination play important roles (Murphree & Breznitz, 2018). Through a state authority specialized in standardization management, i.e., the China National Institution of Standardization, the

Chinese government sets up SSOs, or so-called “technical committees,” to coordinate formal standardization processes.³ Technical committee members are stakeholders sourced from various industry actors such as (state-owned) firms but also universities and government research institutes. Their primary task is to review and amend new standard proposals or standard revisions to reach consensus about adopting a technology as an official industry standard. Even though industry experts’ technical knowledge is highly valued in determining the specific technical contents of standards, government interventions often occur to ensure that certain (domestic) technologies that are preferred by state agencies, such as the TD-SCDMA mobile technology standard, are adopted as technology standards (Gao, 2015; Kwak et al., 2011).⁴

However, the Chinese government is not blind to other industry actors with differing objectives and, hence, increasingly interacts with powerful domestic and international firms or coalitions and other members of formal SSOs to align expectations and coordinate standardization efforts (Liu & Li, 2016; Stewart et al., 2011). With state authorities as relevant actors in coordinating the adoption of a technology as an industry standard in China, firms are expected to leverage their social capital with government institutions to enhance their power in standard competition. In particular, cooperation with Chinese state-owned enterprises is potentially beneficial because these firms are direct government representatives and their R&D efforts “symbolize a direct technology output of the government” (Lei et al., 2012: 238). Hence, in the Chinese context, the relationship between the firm’s network characteristics and the likelihood of setting formal standards is expected to benefit from a firm’s connections with Chinese state-owned enterprises. We therefore propose:

Hypothesis 4 In China, an increasing share of Chinese state-owned enterprises in a firm’s R&D network positively moderates the relationship between the firm’s network characteristics and the likelihood of setting formal standards.

3 Data and methodology

3.1 Data and sample

Since our main objective is to examine how differences in firms’ social relationships affect their involvement in standard setting over time, we take the firm as our unit of analysis and explore yearly firm-level data on the Chinese automotive industry. We collected the data we use between 2012 and 2014, as part of a large-scale nationally funded research project on standardization in the Chinese automotive sector (see also Xu et al., 2017; Dai et al., 2018; Wen et al., 2020; Xu & Zeng, 2021 who use different cross-sections and time periods from similar data sources).

³ The China National Institution of Standardization is a research institute and administrative organization addressing all standardization issues that are relevant for the Chinese economy and society, both nationally and internationally. The organization is directly responsible for *inter alia* the operation of national technical committees, and the administration and exchange of standardization data.

⁴ TD-SCDMA, or Time division-synchronous code division multiple access, is a third-generation mobile telecommunication standard initially developed as a national standard in China.

The data regarding formal standards are systematically retrieved from the China National Knowledge Infrastructure China Standards Database by using a web-crawler.⁵ This database contains worldwide standards records published by Chinese standards organizations starting from the 1970s. The records provide information about various standard characteristics, including *inter alia* title, content, drafters, draft organizations, commencement date, and legal status. We only considered standards in the “automotive” category. Each firm indicated in the standards records as a drafter of a standard is selected and considered an entry in our database. We counted all standards set by each firm per year and over time. In this way, we created an *initial* panel dataset with all 2,811 standards set in the automotive industry from 1985 to 2010. Firms that have set standards but cease to exist in the period under consideration are also included in the (unbalanced) panel, alleviating potential attrition concerns regarding survivorship bias. Because we are also interested in the information on the precise drafting organization(s) responsible for standard setting, the dataset is limited to those standards for which all necessary information is available.

Next, to identify the interorganizational R&D collaborations among Chinese inventors, we use patent data from the Patent Information Services Platform constructed by the Chinese State Intellectual Property Office (SIPO) from 1987 to 2010. All granted patents classified as “automotive” are also retrieved by using a web-crawler.⁶ These patent records provide information about the title, inventors, applicants, times of application and granting, patent classification code, and patent type. The inventors’ data on the patent records are used to identify co-patents, which define the link between firms that collaborate in R&D.

Unfortunately, the patent records do not reveal all necessary information to make an appropriate differentiation by ownership of the patents. While the applicant data reveals whether the owners of the patents are firms, domestic-foreign joint ventures, universities, or research institutes, the records do not allow to make a distinction between Chinese state-owned enterprises or private firms. To do so, firm ownership statistics are retrieved from the Enterprise Credit System provided by the Chinese Administration for Industry and Commerce.

Although patent data have been studied extensively, only a few recent studies work with Chinese patents provided by Chinese State Intellectual Property Office (see Tong et al., 2018). Combined with the China Standards Database and ownership data of firms, a unique panel data setting is created. After merging the standards and patent databases, and including firm-level variables, we cleaned the data for missing observations. The final dataset thus only includes firms for which all necessary information is available from all three main data sources combined. Our resulting *final* sample includes 9287 observations in an unbalanced panel of 508 firms, involved in 1727 standard setting activities in the Chinese automotive industry from 1987 to 2010.⁷ An overview of the number and type of standard setting activities of the firms in our final sample is provided in Table 1. Relative

⁵ The China National Knowledge Infrastructure is a digital library and the largest national electronic repository and publishing house for China based information and resources. The library not only contains a comprehensive collection of Chinese academic and non-academic journals and books, but also data from year-books, population census, patent repositories, and standards administration.

⁶ In our data collection we explicitly considered the time discrepancy between patent applications and grant dates to overcome potential selection bias in the patent data. Following Xu et al. (2017), we applied a four-year time lag and only considered patent data up to and including 2010.

⁷ Note that the incomplete coverage of the ownership data in the Enterprise Credit System substantially limits the total number of observations in our sample. Consequently, the number of standards explored in our sample decreases from 2,811 in the initial sample to 1,727 in the final sample.

Table 1 Sample firms' involvement in standard setting activities

	Firms	Standards	Standards per Sample Firm Type		
			CSOE	DFJV	Private
Total Number of Firms	508		268	48	192
Total Number of Standards		1727	1,205	110	412
Domestic Standards	445	1355	947	92	316
International Standards	115	372	269	18	85
Non-Collaborative Standards	126	544	461	24	59
Collaborative Standards	429	1183	744	86	353
Standards as Principal Editor	233	999	812	35	152
Standards as Non-Principal Editor	348	728	393	75	260

Domestic and International Standards are defined by their level of market approval; (Non-)Collaborative Standards are defined by the number of different actors collaborating in setting the standard, single versus multiple; Standards set as (Non-)Principal Editor are defined by a firm's involvement in setting a collaborative standard, either as principal drafter of the standard or not; CSOEs stands for Chinese state-owned enterprises; DFJVs stands for domestic-foreign joint-ventures; the sample includes 508 firms over 1987–2010

Table 2 Sample firms' involvement in patenting activities

	Patents	Firms
Total	5458	188
Non-Cooperative Patents	5163	180
Cooperative-Patents	295	52
Cooperative-Patent Partners		366
Chinese State-Owned Enterprises		123
Research Institutes		69
Domestic Foreign JVs		32
Private Firms		142

(Non-)Cooperative Patents are defined by the number of different organizations hosting the inventors that filed a patent, single versus multiple; Cooperative Patent Partners represent the number of different cooperative patent partner of the sample firms, subdivided per type; Chinese state-owned enterprises are firms with the Chinese state as principal owner; Research Institutes includes universities, public research institutes, and government research agencies; Domestic Foreign JVs, are joint-ventures between Chinese and foreign firms; Private Firms are privately owned Chinese firms; the sample includes 508 firms over 1987–2010

to private firms, Chinese state-owned enterprises are substantially more involved in setting non-collaborative standards and act relatively more often as principal editor. Furthermore, Table 2 provides an overview of the patenting activities of these firms.⁸

⁸ While the number of co-patents relative to the total number of patents appears to be low, a ratio of approximately 5.4 percent for our time period under consideration is in line with what could be expected (c.f. Belderbos, Cassiman, Faems, Leten, and Van Looy 2014; Briggs and Wade 2014).

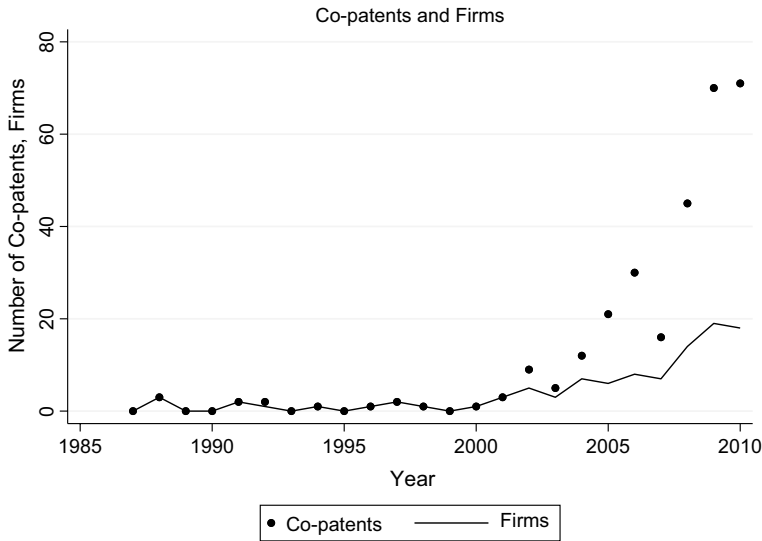


Fig. 1 The distribution of co-patents and firms involved in co-patenting over the sample period. The sample includes 295 co-patents over 1987–2010

Figure 1 shows the distribution of the co-patents and the number of firms involved in co-patenting that are included in our sample over time. The skewed distribution of the sample firms' co-patents shows only limited R&D collaborations until 2000, but rapidly increasing numbers thereafter. Furthermore, the number of firms involved in co-patenting is relative stable up to 2007, but these firms start to collaborate with more different partners over time from 2002.

3.2 Variables and measures

3.2.1 Dependent variables

Our main models use *Total Standards* as the dependent variable, which is measured by the total count of formal standards set by a specific firm in a particular year. Each firm that appears as a drafting organization in the formal standard document receives a count. Note that most standards are collaborative and have multiple drafting firms. The information provided in the documentation allows us to distinguish standards set by drafting firms that act as principal editors. These principal editors are particularly interesting, because they initiate and coordinate the drafting process and benefit more from the actual standard setting relative to co-drafting firms. Therefore, we expect that principal editors are more likely to leverage their collaborative partners to make the standard setting procedure successful. Similarly, we have information on the number of non-collaborative standards set by a single firm. Given that these particular firms cannot make use of the networks of other co-drafters, they are more likely to leverage their own R&D partners for support of their technology as the dominant design. Therefore, we perform sub-sample analyses with the number of *Standards Set as Principal Editor* as well as the number of *Non-Collaborative Standards* as additional dependent variables.

3.2.2 Independent variables

We use three different measures of R&D collaboration based on co-patent data to assess the role of social coordination in firms' involvement in standard setting. Although previous research emphasizes that co-patents are often incomplete measures of R&D collaboration and not all collaborative work efforts are patentable (e.g., Giuri et al., 2007), they do offer the next best solution for quantifying R&D networks if there is a lack of internal information flows about collaboration or social interactions (Goetze, 2010). Using each granted co-patent as a formal collaborative link between firms (c.f., Ardila et al., 2020), we construct both binary and multi-valued adjacency matrices. To deal with the issue that patents are usually filed after the actual R&D collaboration takes place, we follow previous empirical work (e.g., Schilling & Phelps, 2007) and use a three-year rolling window to construct our R&D network time series. Our panel dataset allows us to calculate the properties of R&D collaborations at the firm-level. Hence, we take a firm-centric approach and calculate three basic network measures of social capital (c.f. Borgatti et al., 1998), for each firm-year combination in our panel (see Hanneman & Riddle, 2005 for technical discussions).

To test Hypothesis 1, we use *Direct Ties* which represents a simple network size measure and is defined by the number of other actors to which the firm is directly connected. In our case, the number of other actors with which the focal firm has a granted co-patent in a particular year. For example, in absence of other network changes, if a focal firm has filed a co-patent with two other actors in a particular year, our direct ties measure of this and the next two firm-year combinations has value two.

To capture the strength of a tie to test Hypothesis 2, we use a measure based on the frequency of contact between the focal firm and the other actors it is directly connected to. For each direct tie, the more co-patents filed by the focal firm and the collaborative partner in a year, the stronger the connection. Given that a focal firm can have more collaborative partners in the same year, we define *Tie Strength* as the average number of co-patents of all direct ties of each focal firm in a particular year. A simple illustration: In absence of other network changes, if a focal firm has filed five co-patents with actor A and one co-patent with actor B in a particular year, our tie strength measure of this and the next two firm-year combinations has value three.

Efficient Ties, a measure of efficiency used for testing Hypothesis 3, is defined by the focal firm's number of direct ties minus the average number of direct ties among the collaborative partners themselves, normalized by the actual size of the focal firm's network in a particular year. This measure of network efficiency reflects the overall impact of the network for the focal firm. If each collaborative partner is unique and not reachable through another partner in the network, the focal firm's network is optimally efficient. Suppose for a particular year, the focal firm has direct connections to only three collaborative partners, but these three partners do not have connections among each other. In that case the network of the focal firm is fully efficient, and the efficient ties measure has value one for this firm-year combination. Alternatively, suppose for a particular year that the focal firm is directly connected to three collaborative partners that are also connected to one another. The focal firm can easily reach all other firms through one of its collaborative partners, making the other direct ties 'redundant' or an opportunity cost of social capital. In this example, in absence of other network changes, the efficient ties measure has value one-third for this and the next two firm-year combinations.⁹

⁹ A numerical example for clarification. If a firm has only three collaborative partners, i.e., network size is three, and these partners are not connected to each other. i.e., the average number of direct ties among the partners themselves is 0, the efficient ties measure is equal to $(3-0)/3 = 1$. Alternatively, if all partners are

Provided that firms are restricted by the resources they can invest to build collaborations that may facilitate standard setting, higher network efficiency reflects the focal firm's optimal use of resources to access social capital. Based on the definitions, we expect a negative sign for *Direct Ties* and a positive sign for both *Tie Strength* and *Efficient Ties*.

Finally, we create the *Chinese State-Owned Enterprises* (CSOEs) variable to test Hypothesis 4. This variable is measured yearly, by dividing the number of Chinese state-owned enterprises as collaborative partner of a firm, by the total number of collaborative partners of this firm. If a firm does not have a Chinese state-owned collaborative partner in a year, the value is set to zero. Consequently, the variable is time-variant and normalized for network size.

3.2.3 Controls

Since other Chinese state-related actors, such as research institutes and domestic-foreign joint-ventures, may also direct technology development (Lei et al., 2012) and hence standardization, we control for ownership characteristics of these collaborative partners in firms' R&D networks. Our *Research Institutes* control includes universities, public research institutes, and government research agencies. Similarly, we control for the network presence of *Domestic-Foreign Joint-Ventures* (JVs). While the domestic partners in such joint ventures are not necessarily state-owned, many in our dataset actually are. Both control variables are measured yearly, and in the same way as our Chinese state-owned enterprise independent variable, by dividing the total number of each type of collaborative partner (i.e., research institutes or domestic-foreign joint-ventures) of a firm, divided by the total number of collaborative partners of this focal firm per year.¹⁰ Furthermore, previous work by Soh (2010) reveals that firms with higher innovation intensity have larger incentives to set standards. Since co-patents are already used to construct our network variables, we measure each firm's innovation intensity by its *Non-Cooperative Patents*. This variable is transformed using the Freeman-Tukey method.¹¹

3.3 Descriptive statistics

The descriptive statistics, variance-inflation factors (VIFs), and correlations of the variables are summarized in Table 3. Three aspects stand out. First, there is low correlation between the number of standards and the network variables. This outcome indicates that most standards are set by firms that do not or hardly have any co-patent relationships. For firms that are involved in co-patent networks, these networks are generally small.¹² Besides, the mean and standard deviation of total standards are relatively small,

Footnote 9 (continued)

also connected to any of the others, the partners all have an average of two connections to each other. As a result, the efficient ties measure is equal to $(3-2)/3 = 1/3$.

¹⁰ Private firms are the reference category.

¹¹ Given that the network variables are approximated by a Poisson distribution including many zeros and a small mean, the most appropriate transformation is the Freeman-Tukey method (see Baguley 2012).

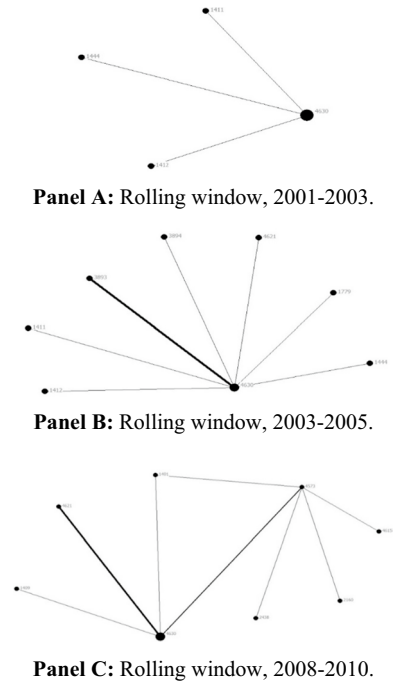
¹² As the descriptive statistics show, many firms in the Chinese automotive industry have no or only a limited number of R&D collaborations. Consequently, the limited variation in the total network data makes social network measures based on the total network, such as closeness and betweenness, difficult to examine and do not yield meaningful results.

Table 3 Descriptive statistics, VIFs, and pairwise correlations

	Mean	SD	Min	Max	VIFs	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 Total Standards	0.186	1.054	0	29															
2 Domestic Standards	0.145	0.864	0	28	0.927														
3 International Standards	0.040	0.396	0	18	0.673	0.365													
4 Principal Editor Standards	0.108	0.843	0	28	0.919	0.871	0.589												
5 Non-Collaborative Standards	0.059	0.582	0	23	0.749	0.749	0.416	0.853											
6 Direct Ties	0.029	0.296	0	9	3.030	0.020	0.021	0.011	0.016	0.016									
7 Tie Strength	0.034	0.409	0	17	2.050	0.050	0.043	0.041	0.048	0.041	0.469								
8 Efficient Ties	0.016	0.123	0	1	3.190	0.054	0.048	0.038	0.044	0.043	0.653	0.632							
9 CSOs	0.006	0.073	0	1	3.450	0.062	0.046	0.069	0.067	0.064	0.399	0.299	0.467						
10 Research Institutes	0.005	0.064	0	1	1.320	0.011	0.008	0.005	-0.001	0.000	0.256	0.283	0.417	0.310					
11 Domestic-Foreign JVs	0.002	0.047	0	1	1.090	0.004	0.004	0.001	-0.002	-0.001	0.102	0.165	0.224	0.106	0.026				
12 Non-Cooperative Patents	1.172	1.061	1	41	1.140	0.085	0.080	0.050	0.043	0.013	0.221	0.206	0.287	0.128	0.156	0.117			
13 Direct Ties×CSOs	0.009	0.138	0	6	3.880	0.027	0.022	0.031	0.032	0.034	0.685	0.272	0.476	0.640	0.168	0.043	0.173		
14 Tie Strength×CSOs	0.009	0.179	0	7	2.690	0.089	0.063	0.101	0.109	0.096	0.290	0.461	0.396	0.612	0.272	0.031	0.075	0.483	
15 Efficient Ties×CSOs	0.004	0.060	0	1	6.890	0.068	0.048	0.079	0.078	0.080	0.454	0.355	0.556	0.833	0.248	0.080	0.134	0.744	0.734

CSOs stands for Chinese state-owned enterprise; JVs stands for joint-ventures; dependent variable for calculating the VIFs is Total Standards; the sample includes 508 firms with 9287 observations covering the period 1987–2010

Fig. 2 R&D network evolution of First Automobile Works Co., Ltd. (2001–2010). The largest nod of the network in each panel represents First Automobile Works; the thicker the lines, the stronger the connection



which reveals that most firms do not regularly set standards. Second, the three network measures are highly correlated. The primary explanation is twofold: the number of the focal firm's co-patent relationships is a central input for all measures and the variation between the network measures is relatively low due to the limited number of co-patent relationships. Consequently, to avoid multicollinearity problems and overfitting of the data, we test each network measure individually. Likewise, fully specified models are not included in the paper as they may yield biased results. Third, despite large Chinese state-owned enterprises controlling a considerable share of the automotive market, these firms have limited collaborative partnerships with other Chinese state-owned enterprises. Consequently, the mean and standard deviation of the Chinese State-Owned Enterprises (CSOEs) variable is low. Fourth, we estimated VIFs from regressions with only one network measure. Note that the VIF score for the interaction Efficient Ties times Chinese state-owned enterprises is above five indicating that multicollinearity becomes an issue when including all interactions simultaneously in our estimations.

To provide additional insights into the evolution of R&D networks in the Chinese automotive industry, we describe the network development of First Automobile Works Group Co., Ltd. as an example. As shown in Fig. 2, First Automobile Works' development path moves from a simple network structure of three bilateral connections in the early 2000s (Panel A) to a more extensive network with (in)direct connections in 2010 (Panel C). In addition, from 2003 onwards (Panels B and C), the company began to develop more co-patents with similar partners in the same year, thus creating stronger

connections. In Panel C, the network becomes less efficient as two of First Automobile Works' R&D partners also start collaborations.

3.4 Model specifications

Because the dependent variables are discrete counts, we apply count data modelling to perform our estimations. Count data models are intrinsically nonlinear and explicitly designed to deal with distributions that include only nonnegative integer values (Cameron & Trivedi, 2013). Given our dependent variable's distributional properties and our data's panel structure, a Poisson panel estimation is an appropriate technique for our modelling strategy. Since the study's main objective is to examine how differences in firms' social relationships affect their involvement in setting standards over time, we focus on exploiting *within* variation of the data by running fixed effects estimations at the firm-level. An additional advantage of firm-level fixed effects models is that we avoid potential omitted variable bias caused by unobserved firm-, sector-, and region-specific time-invariant factors. All time-invariant predictors are controlled for by the fixed effects model, no matter if they are unobserved or explicitly measured.

A general problem of Poisson models is the equidispersion property, which formally implies equality of the mean and variance of the data. As shown by the descriptive statistics in Table 3, the mean and variance of our dependent variables are overdispersed, raising concerns about whether Poisson estimations are appropriate. The common alternative for dealing with overdispersion is negative binomial regression. While negative binomial models seem attractive, they are often not the most preferred. According to Wooldridge (2010), the distributional assumptions of the negative binomial model are rather strict. By contrast, Poisson models are more flexible and provide consistent estimators even when the distribution is misspecified. Furthermore, as noted by Allison and Waterman (2002: 247), the conditional negative binomial fixed-effects model is "not a true fixed effects method" as it does not necessarily control for all stable or time-invariant covariates. Consequently, Wooldridge (2010) and Cameron and Trivedi (2013) state that applying Poisson panel models with robust standard errors yields more robust results than fixed effects negative binomial panel models. Hence, we employ fixed effects panel Poisson models for our estimations.

Applying non-linear panel Poisson models complicates the interpretation of the coefficients of the main and the interaction effects relative to basic linear panel models (for comprehensive discussions see Cox et al., 2009; Wiersema & Bowen, 2009; Bowen, 2012). Most importantly, the estimated coefficients cannot be interpreted as marginal effects. The non-linearity of the model makes that the marginal effect of an explanatory variable varies over its full range, both without and with interaction terms included. While we offer an incidence rate ratio interpretation for the main coefficients of the direct effects (see Cameron & Trivedi, 2013), the interpretation of the interaction effect is more complex. To get a better understanding of the results of our interacting variables, we opt for a visualization of the estimates in the count metric (Cox et al., 2009). Plotting separate curves of the non-linear relationship between $Y_{(i,t)}$ and $X_{1(i,t)}$ at different values of $X_{2(i,t)}$, offers an intuitive solution to interpret the interaction effect over a meaningful range of the data.

Table 4 Firm fixed effects panel Poisson estimations total, domestic, and international standards

	Total Standards					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Direct Ties	−0.075 (0.151)			−0.017 (0.197)		
Tie Strength		0.138** (0.069)			0.046 (0.069)	
Efficient Ties			0.344* (0.189)			0.214 (0.249)
CSOEs	0.339 (0.224)	−0.045 (0.261)	0.036 (0.210)	0.414 (0.355)	−0.373 (0.461)	−0.361 (0.534)
Research Institutes	0.557 (0.453)	0.347 (0.469)	0.353 (0.479)	0.517 (0.488)	0.326 (0.480)	0.430 (0.474)
Domestic-Foreign JVs	0.353 (0.535)	0.297 (0.515)	0.222 (0.538)	0.335 (0.540)	0.391 (0.513)	0.265 (0.543)
Non-Cooperative Patents	0.090*** (0.030)	0.088*** (0.031)	0.090*** (0.031)	0.090*** (0.031)	0.091*** (0.031)	0.090*** (0.031)
Direct Ties × CSOEs				−0.146 (0.394)		
Tie Strength × CSOEs					0.205 (0.163)	
Efficient Ties × CSOEs						0.596 (0.594)
Year Fixed Effects	YES	YES	YES	YES	YES	YES
Observations	9287	9287	9287	9287	9287	9287
Number of Firms	508	508	508	508	508	508
Log-Likelihood	−2776	−2772	−2774	−2776	−2770	−2774

CSOEs stands for Chinese state-owned enterprises; JVs stands for joint-ventures; robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Besides estimating fixed effects models to alleviate omitted variable bias, we also formally address apparent endogeneity concerns, that is, reverse causality between our network variables and standards, by estimating Granger causality (Granger, 1969). Generally, $X_{(i,t)}$ Granger-causes $Y_{(i,t)}$ if a joint test on the lagged values of $X_{(i,t)}$ in an autoregressive model where lagged values of $Y_{(i,t)}$ are also included, indicates that the values for $X_{(i,t)}$ provide significant information about future values of $Y_{(i,t)}$. Given our data's Poisson distribution and the unavailability of non-linear Granger causality tests for panel data, we add nonlinear functions of lags to the basic linear model and perform a joint test on both the linear and nonlinear terms of the lagged endogenous variable (see He et al., 2009). More specifically, we extend the basic linear model for $Y_{(i,t)}$ as a function of lagged $Y_{(i,t)}$ and lagged $X_{(i,t)}$ with lagged $X_{(i,t)}^2$ and perform a joint test on lagged $X_{(i,t)}$ and lagged $X_{(i,t)}^2$. Our estimations are based on a panel vector autoregression model embedded in a generalized method of moments (GMM) framework (see Abrigo & Love, 2016).

4 Results

4.1 Main results

The results from the panel Poisson regressions for *Total Standards* are reported in Table 4. Models 1 to 3 of Table 4 show that only *Tie Strength* and *Efficient Ties* have a direct and positive significant effect on the total number of standards set by a firm, rejecting Hypothesis 1 but corroborating Hypotheses 2 and 3. Specifically, in Model 2, a one unit increase in *Tie Strength* corresponds to multiplying the expected count of total standards by a factor of 1.15.¹³ Furthermore, the interactions between the network variables and Chinese state-owned enterprises in Models 4 to 6, do not show any significant effects. Consequently, Hypothesis 4 is also rejected for the *Total Standards* sample. Finally, the control variable non-cooperative patents shows that partners in collaborative R&D networks with higher innovation intensity are more likely to set (domestic) standards. A finding consistent with previous empirical research.

4.2 Further analyses

4.2.1 Domestic and international standards

To capture the heterogeneity of standards regarding their level of market approval, we subdivide the overall sample into *Domestic Standards* and *International Standards* and reestimate our models. Table 5 shows the results. For setting domestic standards, the direct effects of the network variables in Models 7 to 9 show similar results as for the *Total Standards* sample. Based on the number of observations, this finding is not surprising as most standards set in China are domestic. In contrast to the findings for total and domestic standards, we find the expected negative and significant effect of the number of direct connections on setting *International Standards* in Model 13. However, we do not find that the strength and efficiency of a firm's connections matter for setting international standards. Regarding our hypotheses, our results reveal that the impact of the different properties of the network connections vary over the types of standards set by firms. Although Hypotheses 2 and 3 are corroborated in the total and domestic standards samples, this effect is not verified for international standards. Similarly, we only find support for Hypothesis 1 in the international standards sample.

In the estimation results of Table 5, we only find evidence for Hypothesis 4, that the relationship between a firm's network characteristics and the likelihood of setting formal standards is positively moderated by an increasing share of Chinese state-owned enterprises in the firm's R&D network, in Model 17. Even though the main effect of *Tie Strength* is not significant, this finding is a first indication that stronger ties with Chinese state-owned enterprises as R&D partners might affect the propensity to set standards.

The control variables in Table 5 offer interesting additional findings. We find fragmented evidence for international standards that firms with a larger share of Chinese state-owned

¹³ Incidence rate ratios can be used to interpret the results. For example, the incidence rate ratio for the coefficient of *Tie Strength* in Model 2 of Table 4 is $\text{irr} = e^{0.138} = 1.15$, which means that the expected count of the total standards is multiplied by a factor of 1.15 when tie strength (i.e., the average number of co-patents across all direct ties of the focal firm) increases by one unit.

Table 5 Firm fixed effects panel Poisson estimations total, domestic, and international standards

	Domestic Standards				International Standards							
	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18
Direct Ties	0.051 (0.156)			0.146 (0.177)			-0.417* (0.252)		-1.665 (1.028)			
Tie Strength		0.127* (0.069)			0.073 (0.071)			0.130 (0.130)		-1.076 (0.737)		
Efficient Ties			0.328* (0.177)			0.323 (0.237)			0.216 (0.414)			-1.004 (1.157)
CSOEs	-0.085 (0.427)	-0.311 (0.425)	-0.267 (0.394)	0.080 (0.513)	-0.551 (0.574)	-0.287 (0.604)	1.327*** (0.428)	0.639 (0.392)	0.831** (0.388)	0.767 (0.533)	0.143 (0.635)	-0.217 (0.745)
Research Institutes	0.573 (0.511)	0.507 (0.510)	0.492 (0.514)	0.540 (0.538)	0.504 (0.518)	0.495 (0.524)	0.686 (0.662)	0.082 (0.973)	0.292 (0.846)	1.898* (1.074)	0.916 (0.903)	1.138 (0.879)
Domestic-Foreign JVs	0.357 (0.570)	0.322 (0.550)	0.256 (0.561)	0.323 (0.574)	0.389 (0.553)	0.258 (0.566)	1.175*** (0.275)	1.214*** (0.283)	1.202*** (0.278)	1.175*** (0.276)	1.199*** (0.285)	1.196*** (0.277)
Non-Cooperative Patents	0.109*** (0.033)	0.106*** (0.033)	0.109*** (0.033)	0.108*** (0.033)	0.108*** (0.033)	0.109*** (0.033)	0.048 (0.054)	0.052 (0.059)	0.049 (0.058)	0.047 (0.054)	0.051 (0.057)	0.047 (0.056)
Direct Ties × CSOEs				-0.283 (0.310)						1.920 (1.458)		
Tie Strength × CSOEs					0.142 (0.136)						1.357* (0.766)	
Efficient Ties × CSOEs						0.029 (0.522)						2.406 (1.565)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8292	8292	8292	8292	8292	8292	2090	2090	2090	2090	2090	2090
Number of Firms	445	445	445	445	445	445	115	115	115	115	115	115
Log-Likelihood	-2363	-2360	-2362	-2362	-2359	-2362	-546	-547	-547	-543	-543	-544

CSOEs stands for Chinese state-owned enterprises; JVs stands for joint-ventures; robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

enterprises in their R&D networks are more likely to set standards. Given that the Chinese government oversees the direction of technology development in strategically important sectors such as the automotive industry, Chinese state-owned enterprises are often considered advantaged in standard setting. However, our main results do not support this premise and, hence, are more in line with the findings of Tang et al. (2016), who indicate that the prominence of Chinese state-owned enterprises in high-tech industries is declining. More consistently, we find that having more domestic-foreign joint ventures in a research network positively and significantly affects setting international standards. Although this finding could well be expected, it also shows that foreign firms play a role in setting international standards and thus facilitate international technological upgrading of the Chinese automotive industry. Similar to the main results, R&D partners with higher innovation intensity as measured by non-cooperative patents are more likely to set domestic standards.

4.2.2 (Principal) editors of the standard drafting process

Our data also offers the opportunity to differentiate between firms that are the *Principal Editor* of a collaborative standard and firms that set *Non-Collaborative Standards*. To clarify the importance of this differentiation, note that for collaborative standards, members' contributions to developing a standard and hence, their future earnings potential, differ (Dai et al., 2018). Given the potential commercial benefits of being a standard's principal editor or exclusive drafter, these firms are more likely to invest in R&D relationships with trustworthy and more committed partners (Wen et al., 2020). Therefore, principal editors and non-collaborative drafters are expected to be particularly likely to gain from social coordination and leveraging their R&D partners to set standards. The results in Table 6 corroborate these expectations for *Tie Strength* in Models 20 and 26. These findings support the idea that it is not the sheer size or efficiency of a firm's R&D network that is important in setting commercially attractive standards, but rather the strength of its R&D collaborations. In terms of our hypotheses, these results provide additional support for Hypothesis 2, but reject Hypotheses 1 and 3 for these particular subsets of the data.

Finally, to provide more detail on the relationship between strong ties and standard setting, in Models 23 and 29 of Table 6, we also examine whether this relationship varies with the proportion of Chinese state-owned enterprises in a firm's R&D network. Contrary to the main results, we find a significant positive moderating effect in both models and hence, support for Hypothesis 4. These results imply that the editors of a standard drafting process are particularly likely to set standards when they have stronger connections with their R&D partners, and even more so when they collaborate relatively more with Chinese state-owned enterprises. Interestingly, we also find positive interaction coefficients for *Direct Ties* and *Efficient Ties* in both subsamples presented in Table 6. Even though the direct effects of both variables are not significant, these findings do emphasize that the share of Chinese state-owned enterprises in a firm's collaborative R&D network is a positive moderator. Hence, these findings offer additional evidence supporting Hypothesis 4. To present a more informative interpretation, the interaction effects from Models 23 and 29 are plotted in Fig. 3. The steep upward gradient of the curve in both plots where the proportion of Chinese state-owned enterprises as R&D partners is 1, implies that if a firm's R&D partners are all Chinese state-owned enterprises, there is a positive (and significant) relationship between tie strength and the setting of standards. On the contrary, if a firm's R&D partners are all non-Chinese state-owned enterprises, i.e., the proportion of Chinese state-owned enterprises as R&D partners is 0, the relationship between tie strength and the setting of

Table 6 Firm fixed effects panel Poisson estimations principal editor and non-collaborative standards

	Standards as Principal Editor						Non – Collaborative Standards					
	Model 19	Model 20	Model 21	Model 22	Model 23	Model 24	Model 25	Model 26	Model 27	Model 28	Model 29	Model 30
Direct Ties	− 0.080 (0.173)			− 0.481 (0.342)			0.031 (0.187)			− 0.493 (0.363)		
Tie Strength		0.214*** (0.083)			− 0.250 (0.153)			0.405*** (0.135)			− 0.275 (0.196)	
Efficient Ties			0.157 (0.208)			− 0.316 (0.325)			0.230 (0.267)			− 0.286 (0.333)
CSOEs	0.394 (0.262)	− 0.156 (0.250)	0.224 (0.239)	− 0.147 (0.439)	− 1.165*** (0.284)	− 0.962** (0.481)	0.223 (0.236)	− 0.611* (0.363)	0.077 (0.271)	− 0.622 (0.507)	− 1.860*** (0.529)	− 2.074* (1.235)
Research Institutes	1.087 (0.736)	0.892 (0.766)	0.930 (0.749)	1.308* (0.742)	1.312* (0.759)	1.183 (0.755)	2.178* (1.114)	1.960* (1.113)	2.109* (1.129)	2.309* (1.224)	2.302* (1.391)	2.446* (1.279)
Domestic-Foreign JVs	0.739 (0.815)	0.545 (0.801)	0.645 (0.838)	0.894 (0.810)	0.904 (0.816)	0.836 (0.818)	1.431*** (0.381)	0.805 (0.526)	1.347*** (0.378)	1.634*** (0.427)	1.628*** (0.407)	1.547*** (0.411)
Non-Cooperative Patents	0.138*** (0.047)	0.140*** (0.047)	0.140*** (0.047)	0.138*** (0.046)	0.143*** (0.046)	0.138*** (0.046)	0.096 (0.069)	0.114 (0.071)	0.099 (0.068)	0.090 (0.069)	0.103 (0.071)	0.093 (0.069)
Direct Ties × CSOEs				1.029* (0.597)						1.410** (0.666)		
Tie Strength × CSOEs					0.783*** (0.178)						1.099*** (0.241)	
Efficient Ties × CSOEs						1.833*** (0.653)						2.808** (1.239)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4834	4834	4834	4834	4834	4834	2735	2735	2735	2735	2735	2735
Number of Firms	233	233	233	233	233	233	126	126	126	126	126	126
Log-Likelihood	− 1555	− 1550	− 1555	− 1553	− 1537	− 1550	− 841	− 834	− 841	− 838	− 825	− 837

CSOEs stands for Chinese state-owned enterprises; JVs stands for joint-ventures; robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

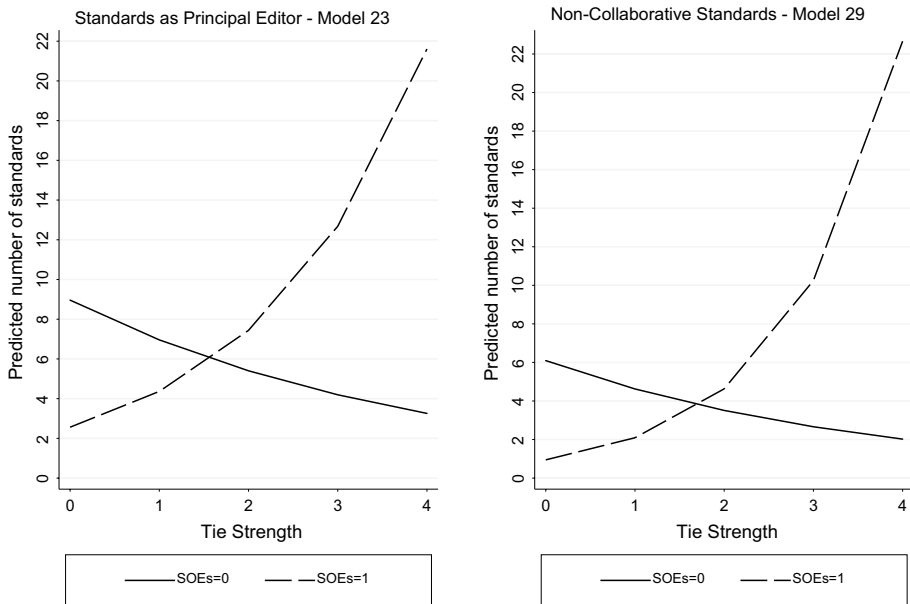


Fig. 3 Interaction plots for the Editors of the standard drafting process subsamples

standards is negative (but insignificant), as showed by the downward sloping curve in both figures.

4.3 Robustness checks

4.3.1 Connected firms

Since, during the time period under consideration, many standards are set by firms without any R&D collaboration, we also ran the same estimations for the subset of firms that do collaborate as a robustness check. These findings are presented in Tables 9 and 10 of the Appendix. Despite the smaller number of observations and increased magnitude of the coefficients, the results are very similar. Again, Hypotheses 2 and 3 are corroborated for *Total* and *Domestic Standards*, but not for *International Standards*. In contrast, support for Hypothesis 1 is only found for the *International Standards* subsample. Support for Hypothesis 4 is absent.

4.3.2 Granger causality

A salient issue in the analysis of networks, collaborations, and alliances is reverse causality. To examine whether our different network properties can be used to predict the number of standards set in the Chinese automotive industry and eliminate concerns about the direction of the relationship, we present the outcomes of our Granger-causality Wald-tests in

Table 7 Panel vector autoregression estimations to test Granger-causality

Dependents [Independents]	Direct Ties		Tie Strength		Efficient Ties	
	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
<i>L1</i>						
Total Standards	0.267	[0.813]	0.063*	[0.343]	0.264	[0.349]
Domestic Standards	0.133	[0.366]	0.156	[0.428]	0.066*	[0.338]
International Standards	0.126	[0.481]	0.254	[0.233]	0.514	[0.783]
Standards as Principal Editor	0.238	[0.657]	0.130	[0.324]	0.221	[0.341]
Non-Collaborative Standards	0.750	[0.626]	0.411	[0.440]	0.932	[0.623]
<i>L2</i>						
Total Standards	0.115	[0.445]	0.039**	[0.369]	0.095*	[0.143]
Domestic Standards	0.061*	[0.334]	0.105	[0.326]	0.004***	[0.186]
International Standards	0.181	[0.374]	0.379	[0.123]	0.587	[0.362]
Standards as Principal Editor	0.174	[0.370]	0.075*	[0.404]	0.090*	[0.179]
Non-Collaborative Standards	0.429	[0.371]	0.465	[0.383]	0.694	[0.331]
<i>L3</i>						
Total Standards	0.074*	[0.414]	0.093*	[0.166]	0.076*	[0.104]
Domestic Standards	0.028**	[0.300]	0.042**	[0.189]	0.015**	[0.065*]
International Standards	0.647	[0.346]	0.655	[0.322]	0.638	[0.207]
Standards as Principal Editor	0.135	[0.191]	0.142	[0.214]	0.082*	[0.124]
Non-Collaborative Standards	0.268	[0.617]	0.469	[0.358]	0.460	[0.383]

All variables in first differences; L1-L3 denote the number of lags of $Y_{(i,t)}$ and $X_{(i,t)}$ included in the model; numbers are p -values of a joint test (Wald) on both the lagged linear and lagged squared terms of $X_{(i,t)}$, conditional on lagged values of $Y_{(i,t)}$; control variables as specified in the baseline regressions are included in the model; panel vector autoregressions satisfy the stability condition in all estimations; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7.¹⁴ These outcomes show the p -values of a joint Wald test with the null hypothesis that the coefficients on all the lags of the linear and squared terms of $X_{(i,t)}$, conditional on lagged values of $Y_{(i,t)}$, are jointly equal to zero. For example, the p -values in Column 1 can be interpreted as whether changes in the number of the focal firm's direct ties predict changes in the number of standards set for different numbers of lags. Column 2 presents the p -values of a joint Wald test on the reverse case and shows whether different types of standards set predict the number of a firm's direct ties. The overall implications of these findings are that, when significant, the direction of the relationship is as expected and runs from the network variables to the different types of standards. The only exception is the prediction for the relationship between efficient ties and setting domestic standards. When including three lags, this particular result in Columns 5 and 6 shows that the direction of the relationship runs both ways. Overall, the findings provide no indications that reverse causality might be an issue, and we only identified one case of simultaneity. We therefore

¹⁴ Note that we only use Granger-causality to check whether our different R&D network properties predict the number of standards set by a firm (and rule out that the effect runs the other way around). We do not infer causality from these results (given that other potential causes cannot be ruled out, this is also not possible for these specifications).

Table 8 Overview of the hypothesis support per dependent variable

	Hypothesis 1	Hypothesis 2	Hypothesis 3	Hypothesis 4
Total Standards		✓	✓	
Domestic Standards		✓	✓	
International Standards	✓			
Standards as Principal Editor		✓		✓
Non-Collaborative Standards		✓		✓

conclude that our network variables can be used to predict the number of standards set by firms in our setting.

Generally, our results indicate that the impact of R&D connections on setting standards varies over the properties of the connections and the type of standards set. Consequently, we find support for our hypotheses, but this support is conditional upon the heterogeneity of standard types.¹⁵

5 Discussion, conclusions and implications

In this study, we leverage a unique panel dataset to examine how firms' networks of R&D collaboration and the properties thereof can enhance their propensity to set formal standards in the Chinese automotive industry. We do not find full support for all hypotheses in the main results exploiting the full data sample, but we obtain additional support for our hypotheses by examining subsets of the data. Table 8 presents a comprehensive overview of the support for each hypothesis per dependent variable used in the estimations. Using the insights from Table 8, our findings provide two novel contributions to the literature on standardization and innovation management.

First, our results present new evidence for the premise that building social capital among R&D network partners is an important determinant of standard setting success. In particular, from a focal firm's perspective, we conclude that building social capital by establishing strong relationships among R&D network partners helps lobbying and aligns stakeholders to reach consensus on the specifications and adoption of a formal standard in technical committees. Because collaborative R&D efforts involve sharing tacit and proprietary knowledge to create new technologies (McFadyen et al., 2009), levels of trust, reciprocity, and common objectives between R&D partners are generally high. Leveraging social capital through coordination thus explains why strong R&D connections facilitate a focal firm's effort to set standards, while having strong connections to business partners with limited social capital does not (c.f. Dai et al., 2018). Exploitation of social capital among R&D network connections is thus an essential source of competitive advantage in formal standard competition in the Chinese automotive industry. Although specific to R&D collaborations, our findings are consistent with the work of Bar and Leiponen (2014) and underline that, in technology intensive relationships, firms tend to leverage preexisting connections to lobby in standard setting procedures.

¹⁵ Further analyses and robustness checks are included in the supplementary material available online.

Second, by exploiting the details of the Chinese standards records, we show that the impact of R&D connections and their different properties varies with the type of standards set. While previous empirical findings indicate that having more connections represents an opportunity cost of social capital and hence has a negative effect on setting standards (Bar & Leiponen, 2014), we show that this only holds for setting international standards, not for the domestic or total standards sample. An intuitive explanation is that, relative to total or domestic standards, the opportunity costs of social capital are higher for building the necessary (international) relationships to support setting international standards. In contrast, the proposed positive effects of strong and efficient R&D connections only hold for total and domestic standards, not international ones. Furthermore, stakeholders' contributions to drafting standards also matter. Principal editors of collaborative standards and non-collaborative standard setters benefit most from social coordination of R&D partners with which they have strong connections. Moreover, they particularly benefit from their connections when the proportion of Chinese state-owned enterprises in their R&D network increases. This effect can be explained by the overrepresentation of Chinese state-owned enterprises in both subsamples. For Chinese state-owned enterprises it is probably easier to reach out and build social capital with other Chinese state-owned enterprises than with private firms. Clearly, for the total, domestic, and international standards samples no relative overrepresentation of Chinese state-owned enterprises exist. Hence, the firms in these samples do not benefit from an increased proportion of Chinese state-owned enterprises in their R&D network. Overall, considering the different types of standards provides new insights into the heterogeneity of standards and the fact that standards differ in technological contribution and market value (c.f. Dai et al., 2018; Wen et al., 2020).

Overall, the impact of the different type of interorganizational connections between R&D network partners on standard setting is clearly not uniform. The positive relation between having strong R&D connections and standard setting is corroborated for all but the international standards sample. In contrast, the negative impact of the absolute number of different connections is only relevant for setting international standards, while the efficiency of the connections holds for the total and domestic standards samples. These findings have important implications for scholars, as exploring the inherent heterogeneity in the type of interorganizational connections may prove essential for advancing our understanding of how the social coordination of innovation activities may affect standardization performance.

Our findings also have important managerial implications and policy relevance. For managers, our results imply that particularly strong and efficient collaborative partnerships are not only useful for enhancing technological knowhow or increasing a firm's innovative capabilities, but they are also essential strategic assets that provide a competitive edge in standard competition. Managers should accommodate technological cooperation and build social capital with other organizations and, concurrently, be aware of the power of these partners in the industry. In addition, it is important to note that developing a network of collaborative activities takes time, and it is often a complex task for managers to find the right technological partners (Nieto & Santamaría, 2007). Therefore, prior to effectively leveraging the power of a firm's network position to set standards, managers need to implement long-term strategies to intensify existing collaborations and build their R&D networks over time.

For policy makers, it is important to realize that social coordination with the intent to set standards does not always lead to optimal outcomes. Firms with well-developed R&D network positions are likely to start lobbying with their collaborative partners to set the standard alliances from the onset of new technology development. Given the competitive advantage and potential monopoly rents that firms can gain from setting a standard, the market might run the risk that the best lobbied standard becomes adopted, not the most technologically advanced. The best-known Chinese example is the adoption of the (inferior) TD-SCDMA mobile technology standard (see Gao, 2015). As argued by Murphree and Breznitz (2018), most major controversial standards in China involve state actors, such as Chinese state-owned enterprises, universities, and government (research) institutes. Given the state's dominant role in the automotive industry, policy makers should also be aware of the potential negative implications of social coordination and lobbying.

This research is not without its limitations and corresponding future research opportunities. Given the scope of our results, the outcomes are likely to be industry specific. Network characteristics of other industries might be different and consequently, yield different outcomes. To assess the general applicability of our results, future research efforts could be directed to increasing our understanding of firm- and sector-level heterogeneity in explaining the ability to set standards. From a firm-level perspective, it would also be interesting to further unpack the black box of social coordination. The role of our theoretical constructs e.g., common interest, trust, and aligned objectives could be tested directly in the context of standard setting. We also took a limited view of R&D collaborations, especially concerning the nature of the relationships between the R&D partners. This nature is important, because competing firms may behave strategically in the formation of alliances in SSOs and hence affect the number of standards particular firms set (Axelrod et al., 1995; Sakakibara, 2002). Yet, a firm's network consists of many more connections, both formal and informal, that may play a role in standard setting, especially in contexts that are institutionally less developed. From a sector-level perspective it would be useful to know whether the importance of leveraging relationships differs across sectors and why firms in some sectors are more prone to use social coordination for standard setting than others. Still, not all relationships in a social network are equally valuable. Understanding and empirically testing which relationships contribute most to a firm's competitive ability to set standards would open a new domain of research opportunities. From a country-level perspective, it would be interesting to explore whether our results are generalizable to other emerging economies hosting relatively large and open automotive industries with significant government intervention, such as Mexico and Brazil. The importance of cultivating R&D collaborations and its implications for setting (international) standards might well be applicable to these countries.

Finally, this research domain is particularly interesting when considering the newcomers in the Chinese automotive industry. Around 2014, a group of innovative start-ups emerged that focus on the development of smart electric vehicles for technology-savvy middle-class consumers, including NIO and Xiaopeng. These new ventures, founded by private actors, are changing the traditional Chinese automotive sector into a high-tech industry, including innovative IT and artificial intelligence applications. So far, these newcomers are gaining market share but do not yet replace the traditional automotive companies that for long dominate the market. Nevertheless, with their high-tech orientation, they have become active players in R&D and standardization. Exploring the role of these newcomers in driving R&D collaborations and standard setting could provide valuable new insights in whether their collaborative activities and performance differ from the traditional automotive companies analyzed in this study.

Appendix

See Tables 9 and 10

Table 9 Firm fixed effects panel Poisson estimations of connected firms only

Connected Firms	Total Standards					
	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18
Direct Ties	−0.013 (0.150)			0.154 (0.194)		
Tie Strength		0.180** (0.084)			0.190* (0.099)	
Efficient Ties			0.743** (0.325)			0.690* (0.375)
CSOEs	0.266 (0.360)	−0.250 (0.414)	−0.296 (0.397)	0.545 (0.468)	−0.216 (0.511)	−0.471 (0.833)
Research Institutes	0.950* (0.488)	0.815* (0.424)	0.655 (0.490)	0.840 (0.531)	0.820* (0.432)	0.680 (0.501)
Domestic-Foreign JVs	0.415 (0.626)	0.416 (0.601)	0.196 (0.633)	0.363 (0.631)	0.405 (0.610)	0.210 (0.646)
Non-Cooperative Patents	0.023 (0.050)	0.013 (0.047)	0.013 (0.050)	0.019 (0.049)	0.012 (0.050)	0.014 (0.051)
Direct Ties × CSOEs				−0.469 (0.372)		
Tie Strength × CSOEs					−0.021 (0.143)	
Efficient Ties × CSOEs						0.262 (0.877)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1009	1009	1009	1009	1009	1009
Number of Firms	55	55	55	55	55	55
Log-Likelihood	−392	−388	−387	−391	−388	−387

CSOEs stands for Chinese state-owned enterprises; JVs stands for joint-ventures; robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 10 Firm fixed effects panel Poisson estimations of connected firms only

Connected Firms	Domestic Standards					International Standards							
	Model 19	Model 20	Model 21	Model 22	Model 23	Model 24	Model 25	Model 26	Model 27	Model 28	Model 29	Model 30	
Direct Ties	0.136 (0.171)			0.252 (0.235)			-1.054* (0.561)			-0.924 (0.779)			
Tie Strength		0.195** (0.092)			0.190* (0.102)			-0.072 (0.262)			-0.522 (1.020)		
Efficient Ties			0.748** (0.305)			0.751** (0.377)			0.096 (0.945)			0.245 (1.505)	
CSOEs	-0.002 (0.541)	-0.350 (0.546)	-0.382 (0.530)	0.210 (0.644)	-0.372 (0.587)	-0.370 (0.919)	1.152** (0.530)	0.244 (0.917)	-0.060 (0.958)	1.320 (1.277)	0.162 (0.905)	0.121 (1.472)	
Research Institutes	0.738 (0.513)	0.720 (0.450)	0.527 (0.479)	0.702 (0.559)	0.718 (0.452)	0.525 (0.504)	3.067*** (1.064)	2.843* (1.536)	2.650*** (1.259)	2.970*** (1.241)	3.053* (1.593)	2.581** (1.259)	
Domestic-Foreign JVs	0.468 (0.622)	0.463 (0.580)	0.256 (0.597)	0.432 (0.625)	0.471 (0.590)	0.255 (0.609)	2.136* (1.198)	2.240* (1.200)	2.270* (1.202)	2.135* (1.199)	2.195* (1.224)	2.276* (1.205)	
Non-Cooperative Patents	0.060 (0.065)	0.046 (0.059)	0.050 (0.064)	0.056 (0.065)	0.047 (0.062)	0.049 (0.065)	-0.199 (0.141)	-0.195 (0.151)	-0.189 (0.136)	-0.204 (0.138)	-0.187 (0.150)	-0.193 (0.134)	
Direct Ties×CSOEs				-0.355 (0.401)						-0.324 (1.933)			
Tie Strength×CSOEs					0.013 (0.134)						0.485 (0.999)		
Efficient Ties×CSOEs						-0.019 (0.950)						-0.355 (2.532)	
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Observations	975	975	975	975	975	975	284	284	284	284	284	284	
Number of Firms	53	53	53	53	53	53	13	13	13	13	13	13	
Log-Likelihood	-365	-360	-361	-364	-360	-361	-78.0	-80.6	-80.7	-77.9	-80.5	-80.7	

CSOEs stands for Chinese state-owned enterprises; JVs stands for joint-ventures; robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

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